공학수학(2)

12.1 Vector Functions and Space Curves

Vector-Valued Functions

In general, a function is a rule that assigns to each element in the domain an element in the range.

A **vector-valued function**, or **vector function**, is simply a function whose domain is a set of real numbers and whose range is a set of vectors.

We are most interested in vector functions **r** whose values are three-dimensional vectors.

This means that for every number t in the domain of **r** there is a unique vector in V_3 denoted by $\mathbf{r}(t)$.

Vector-Valued Functions

If $f(t)$, $g(t)$, and $h(t)$ are the components of the vector $\mathbf{r}(t)$, then f, g, and h are real-valued functions called the **component functions** of $\mathbf{r}$ and we can write

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

We use the letter t to denote the independent variable because it represents time in most applications of vector functions.

Example 1

If $\mathbf{r}(t) = \langle t^3, \ln(3-t), \sqrt{t} \rangle$

then the component functions are

$$f(t) = t^3 \quad g(t) = \ln(3-t) \quad h(t) = \sqrt{t}$$

By our usual convention, the domain of **r** consists of all values of *t* for which the expression for **r**(*t*) is defined.

The expressions t^3, ln(3 − *t*), and $\sqrt{t}$ are all defined when 3 − *t* > 0 and *t* ≥ 0.

Therefore the domain of **r** is the interval [0, 3).

Limits and Continuity

The **limit** of a vector function **r** is defined by taking the limits of its component functions as follows.

1 If $\mathbf{r}(t) = \langle f(t),\ g(t), h(t) \rangle$, then

$$\lim_{t \to a} \mathbf{r}(t) = \left\langle \lim_{t \to a} f(t),\ \lim_{t \to a} g(t), \lim_{t \to a} h(t) \right\rangle$$

provided the limits of the component functions exist.

Limits of vector functions obey the same rules as limits of real-valued functions.

Example 2

Find $\lim_{t\to 0}\mathbf{r}(t)$, where $\mathbf{r}(t) = (1+t^3)\mathbf{i} + te^{-t}\mathbf{j} + \dfrac{\sin t}{t}\mathbf{k}$.

Solution:

According to Definition 1, the limit of **r** is the vector whose components are the limits of the component functions of **r**:

$$\lim_{t\to 0}\mathbf{r}(t) = \left[\lim_{t\to 0}(1+t^3)\right]\mathbf{i} + \left[\lim_{t\to 0} te^{-t}\right]\mathbf{j} + \left[\lim_{t\to 0}\frac{\sin t}{t}\right]\mathbf{k}$$

$$= \mathbf{i} + \mathbf{k}$$

Limits and Continuity

A vector function **r** is **continuous at *a*** if

$$\lim_{t \to a} \mathbf{r}(t) = \mathbf{r}(a)$$

In view of Definition 1, we see that **r** is continuous at *a* if and only if its component functions *f*, *g*, and *h* are continuous at *a*.

Space Curves

There is a close connection between continuous vector functions and space curves.

Suppose that *f*, *g*, and *h* are continuous real-valued functions on an interval *I*.

Then the set *C* of all points (*x*, *y*, *z*) in space, where

$$\mathbf{2} \quad x = f(t) \quad y = g(t) \quad z = h(t)$$

and *t* varies throughout the interval *I*, is called a **space curve**.

Space Curves

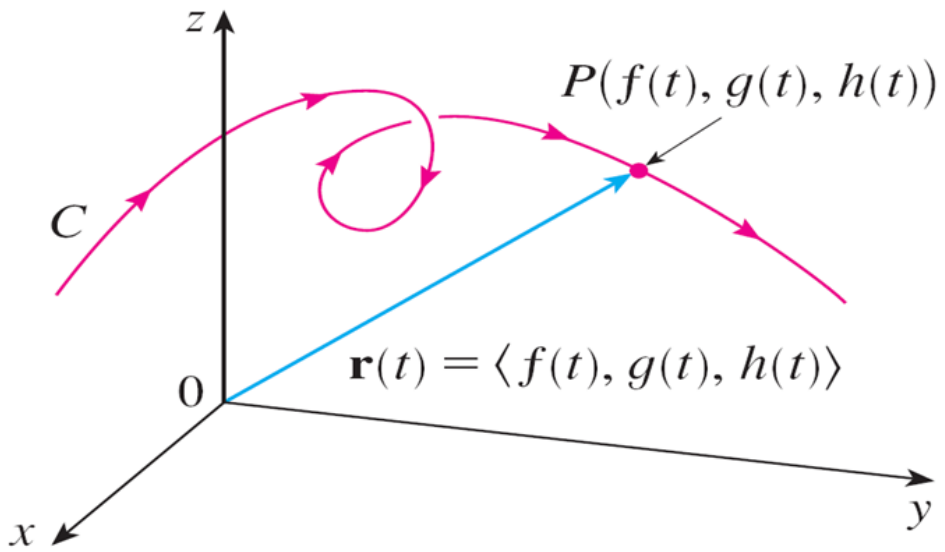
The equations in (2) are called **parametric equations of *C*** and *t* is called a **parameter**.

We can think of *C* as being traced out by a moving particle whose position at time *t* is $(f(t), g(t), h(t))$.

If we now consider the vector function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then $\mathbf{r}(t)$ is the position vector of the point $P(f(t), g(t), h(t))$ on *C*.

Space Curves

Thus any continuous vector function **r** defines a space curve *C* that is traced out by the tip of the moving vector **r**(*t*), as shown in Figure 1.



C is traced out by the tip of a moving position vector **r**(*t*).

Figure 1

Example 4

Sketch the curve whose vector equation is

$$\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\,\mathbf{k}$$

Solution:

The parametric equations for this curve are

$$x = \cos t \quad y = \sin t \quad z = t$$

Since $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$, the curve must lie on the circular cylinder $x^2 + y^2 = 1$.

The point (x, y, z) lies directly above the point $(x, y, 0)$, which moves counterclockwise around the circle $x^2 + y^2 = 1$ in the xy-plane.

Example 4 – Solution

(The projection of the curve onto the *xy*-plane has vector equation $\mathbf{r}(t) = \langle \cos t, \sin t, 0 \rangle$.) Since $z = t$, the curve spirals upward around the cylinder as t increases. The curve, shown in Figure 2, is called a **helix**.

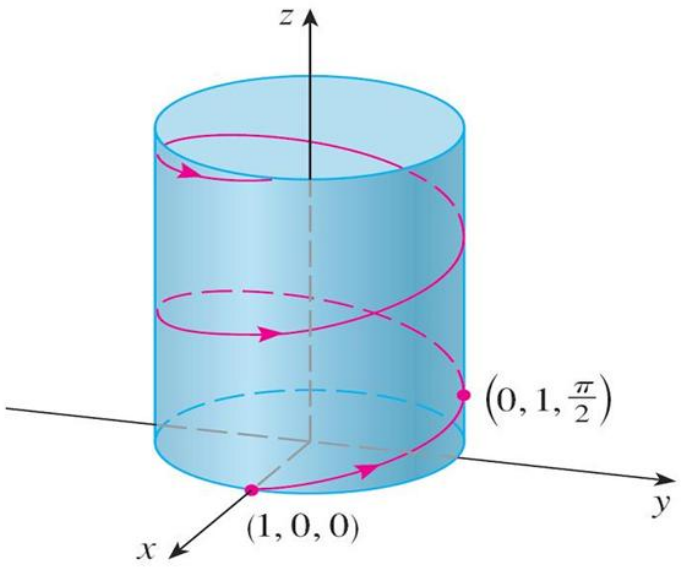


Figure 2